

Forms of Energy – New/alternative Forms of Energy

Different Kinds of Energy

- Fossil fuels
 - Coal
 - Natural gas
 - Petroleum
- Nuclear Energy
- Renewable Energy
 - Wind energy
 - Hydro energy
 - Solar Energy
 - Biomass
 - Geothermal
 - Tidal Power

What is renewable Energy

Renewable Energy describes a collection of energy technologies, that are never-ending and can be replenished.

For example, oil is not a renewable energy technology, because it utilizes limited resources, that will ultimately deplete.

Advantages of renewable Energy

- It is Eco-Friendly
- It is a renewable Resource
- It is a reliable source of Energy
- Leads to Job creation
- Less Maintenance
- Boosts Public Health

Disadvantages of renewable Energy

- Electricity generation is still not large enough
- Low efficiency levels
- Takes a lot of places to install
- Expensive Storage Cost
- Still generates Pollution

Wind energy

Wind makes the blades of the wind turbines turn (like a windmill). These blades are connected to a shaft that turns a generator and produces Electricity.

Wind turbines are easy to put up, so wind farms often consist out of hundreds of them.

Most common areas for Windfarms are:

- Coastal areas
- At the top of rounded hills
- Open plains and gaps in mountains
- Places where wind is strong and reliable

Solar Power

This involves turning sunlight into energy that we can use. They can be built into solar farms which are interconnected solar panels positioned together over many acres.

There are two basic types of solar power:

Photovoltaic energy

Sunlight is converted into electricity by using photovoltaic cells or solar panels.

Solar thermal power

The sun heats up water which is stored in dark containers and this heat is used to generate steam, which drives a turbine.

How is Solar energy stored?

Solar energy can also be stored in electrochemical batteries. When solar energy is pumped into a battery, a chemical reaction among the battery components stores the energy.

Hydropower

Flowing water is collected and turned into electricity. It depends on the amount of water moving. That means how far the water drops and how much it flows through the system. The water is flowing through large pipes which drives turbines (generators which convert energy into electricity).

Tidal Power

Is a special form of Hydropower. A dam is built across a river estuary (Mündungsgebiet). When the tide goes in and out, the water flows through tunnels in the dam. This flow can be used to power a turbine.

Geothermal Energy

Geothermal energy is capturing the earth's interior heat. The heat within 10.000 meters of earth's surface contains 50.000 times more energy than all the oil and natural gas resources in the world.

Some geothermal power stations use steam to turn turbines. Another method would be to use this heat to let a second liquid boil at a lower temperature, so it is more easily converted into steam to run the turbine.

Countries with great geothermal reserves are Iceland, Indonesia, Philippines or Japan.

Biomass

Biomass refers to organic material such as timber and crops grown specifically to be burnt to generate heat and power. It is sustainable and carbon neutral. Most popular biomass are specifically grown crops, agricultural by-products (straw, grain husks), waste wood and animal waste (slurry and chicken litter)

Biofuels

Biofuel is any fuel derived from biomass. These Biomass fuels such as biodiesel and ethanol can be burned in an internal combustion engine like cars or boilers.

Related Links

<https://www.conserve-energy-future.com/advantages-and-disadvantages-of-renewable-energy.php>

<http://www.darvill.clara.net/altenerg/wind.htm>

<https://www.nationalgrid.com/stories/energy-explained/how-does-solar-power-work>

<https://www.aurorasolar.com/blog/storing-solar-energy-everything-you-need-to-know/>